# Proposal for China Commerce Holdings Group's Investment and Participation in the Establishing A Computing Power Platform for the Orchard Land-Photovoltaic Project in Yichuan County, Yan'an City, Shaanxi Province

I. Project Background

(I) Proposal to the Chinese People's Political Consultative Conference

In 2023, Dong Jin, Director of the Coal Mine Cultural Troupe and a member of the Chinese People's Political Consultative Conference, submitted a proposal titled 'Promoting the Fruit Tree + Photovoltaic Model to Support Rural Revitalization,' explicitly proposing to use dwarf-stemmed densely planted apples as the core and to promote the integration and application of cadmium telluride and other thin-film photovoltaic technologies in the forestry and fruit industry;

II) Yan'an Demonstration

In the same year, based on the 300-mu orchard land-photovoltaic demonstration project achieving an average output value of 48,000 yuan per mu and an annual return of over 18%, the governments of Yichuan County and Yan'an City attached great importance to the project and planned to promote it for another 3,000 mu with an installed capacity of 150MWp, adopting an integrated model of "dwarf stock densely planted orchard + power generation glass + microgrid + collective dividends", and listed it as a key support project;

(III) Project approval by the Development and Reform Commission

On 21 September 2024, the Shaanxi Provincial Development and Reform Commission issued the Notice on Issuing the Three-Year Action Plan for the Development of Renewable Energy in Shaanxi Province (2024-2026) to build a 10 GW agricultural-photovoltaic renewable energy base in Weibei;

(IV) Key project of the Ministry of Science and Technology

In the same year, the Yichuan Orchard Land-Photovoltaic Project, as a flagship scientific research cooperation project jointly implemented by China and Europe, was chosen as a key project of the Ministry of Science and Technology's "Intergovernmental International Science and Technology Innovation Cooperation". The project clearly proposed to promote the transformation of agricultural-photovoltaic project from "project demonstration" to "policy system" on the basis of spectral agriculture, thin-film power generation and collective benefit-sharing mechanism.



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*Signature:*

*Date: 20 July 2025*

II. Project Overview

In conjunction with a number of edge computing, data services and AI companies, systematically promoting the design of an integrated solution of "computing power + agriculture + energy", and planning to build an edge computing platform for agricultural production, green energy dispatching and AI intelligent systems. The project adheres to the principle of "agriculture-led, energy-coordinated, and digital-driven". Agricultural enterprises take the lead in proposing scenario requirements, the energy partner provides low-carbon energy supply guarantees, and the digital partner builds intelligent operation and maintenance and AI models to achieve a deep integration of power system dispatching and agricultural digital capabilities. The project will effectively leverage the coordinated implementation of the "dual carbon strategy" and rural revitalization policies, and form a sustainable and scalable integrated solution of "technology-finance-policy".

The project has been chosen as a key project of the Ministry of Science and Technology's "Intergovernmental International Science and Technology Innovation Cooperation", and plans to apply for the "rural comprehensive reform pilot" of the Ministry of Finance, and will be the main applicant for the national agricultural industrial park of the Ministry of Agriculture and Rural Affairs. At the same time, the project is actively striving to be included in the "East Data West Computing" Shaanxi hub node construction plan led by the National Development and Reform Commission.

III. Project Progress

(I) Implementation of the Orchard Land-Photovoltaic Pilot Project

Since 1 May 2023, this project has chosen a 300-mu orchard in Baoding Village, Yichuan County for this proposal experiment. 60,000 dwarf rootstock seedlings were planted, and the experimental results showed the best growth performance despite being the latest planted in the county; 50% of the area of cadmium telluride glass was laid per acre, plus the glass transmittance of 20%, the average light gain per acre was 69%, and the experimental results indicated the optimal light ratio for dwarf rootstock dense planting; the average installed power generation capacity per acre was 52KWp, and the experimental results were equivalent to centralized photovoltaics. The preliminary test was an overall success. In February 2024, China Commerce Puhui Holding Group held talks with relevant leaders of Yan'an City and preliminarily confirmed that the 10,000-mu large-scale implementation site of this project would be Yunyan Town, Yichuan County.

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(II) Large-scale implementation of a computing power centre for the orchard land-photovoltaic project

Based on the comprehensive success of the orchard land-photovoltaic pilot project, it is planned to invest in the development of a computing power project for the orchard land-photovoltaic project in Yichuan County, Yan'an City, Shaanxi Province. The total investment in the first phase of the project is CNY 7.5 billion, covering an area of 10,000 mu; 2 million dwarf seedlings will be introduced; 600MWp of cadmium telluride photovoltaic modules will be laid (1.5 billion); 500MW-class ultra-large size (1215×2300) thin-film power generation glass production line will be built (1 billion); 150MW-class computing power centre will be built (4.5 billion), and equipped with a 300MW power, 3-hour capacity 900MWh energy storage system (500 million), so as to promote the high-quality development of the local economy.

(III) Implementation recommendations

After several rounds of preliminary research and coordination, it is currently preliminarily proposed that Government of Yan'an City guide the platform company under China Commerce Puhui Holding Group -- Beijing Jinglong Nongguang Technology Co., Ltd. to collaborate with the distributed photovoltaic aggregation platform under LongShine Technology Group Co., Ltd. -- Xinyao Energy Technology Co., Ltd., and the 'East Data West Computing' data centre service platform jointly established by Isoftstone Information Technology (Group) Co., Ltd. and CNISPGROUP Co., Ltd. — Beijing Isoftstone Cloud Services Information Technology Co., Ltd., to jointly establish the project platform company 'Yichuan County Zhongwa Yuanguang Digital Technology Co., Ltd.'

In terms of project funding arrangements, it is planned that Yichuan Orchard Land-Photovoltaic Investment Management Co., Ltd., a special fund platform established by China Commerce Puhui Holdings Group, will invest as the main fund contributor. Moving forward, in combination with the functional roles of all parties and the progress of the project, the cooperative framework and equity distribution mechanisms will be further clarified.

IV. Investment plan and policy list

(I) Introduction to China Commerce Puhui, Jinglong Nongguang, and Yichuan Investment

China Commerce Puhui Holding Group was formerly known as China Commerce Silk Road Holding Group. It was founded under the authorization of China Commerce Holdings Group and renamed in February 2022. In view of the fact that this project involves specific local matters such as land transfer and photovoltaic quotas, China Commerce Puhui has been established by the county government where the project is located as a guiding platform for orchard land-photovoltaic industry, with clear local implementation attributes.



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The establishment of this platform is the result of the joint efforts of the Yan'an City and Yichuan County governments, and also reflects the local government's high level of attention and institutional support for the orchard land-photovoltaic project. As the project mechanism matures, more qualified district and county governments will be included in the platform system in the future to jointly promote the regional replication of the orchard land-photovoltaic model.

China Commerce Puhui established an industrial platform Beijing Jinglong Nongguang Technology Co., Ltd. (referred to as Jinglong Nongguang), and its shareholder Advanced Solar Power (Hangzhou) Inc. is a world-leading company in the research and development and production of cadmium telluride thin film technology. The pilot project confirmed that cadmium telluride spectrum regulation promotes apple growth.

China Commerce Puhui established an investment platform Yichuan Orchard Land- Photovoltaic Investment Management Company (referred to as Yichuan Investment), whose shareholders are China Development Fund Co., Ltd. and the management team, Beijing Jinlongli Investment Management Centre (Limited Partnership). It will raise and manage funds related to the computing power industry for the orchard land- photovoltaic project.

(II) Development Entity

It is planned that Jinglong Nongguang, Xinyao Energy Technology Co., Ltd., and Beijing Isoftstone Cloud Services Information Technology Co., Ltd. will establish a project platform company Yichuan County Zhongwa Orchard Land-Photovoltaic Digital Technology Co., Ltd. (referred to as Zhongwa Digital Technology, subject to the approval of industrial and commercial registration), with a registered capital of 100 million yuan. Among them, Jinglong Nongguang will invest 51 million yuan and held 51% of the shares, Xinyao Energy will invest 25 million yuan and held 25% of the shares, and Isoftstone Cloud Services will invest 24 million yuan and held 24% of the shares. The detailed terms and conditions will be agreed in the joint venture agreement.

China Commerce Puhui Holding Group
Jinlongli Enterprise Centre (60%) Yichuan State-owned Assets Supervision and Administration Commission (40%)

Beijing Jinglong Nongguang Technology Co., Ltd.
Zhongshang Puhui (91%) Advanced Solar Power (9%)

Xinyao Energy Technology Co., Ltd. Beijing Isoftstone Cloud Services Information Technology Co., Ltd.
Bangdao Tech (89.67%) Zhifeng Partnership (10.33%) Isoftstone (20%) CNISP Honeycomb (9%) CNISP Weifu (9%) China Soft Partnership (20%)

Yichuan Zhongwa Orchard Land- Photovoltaic Digital Technology Co., Ltd.
Zhongshang Huipu (51%) Xinyao Energy (25%) Isoftstone Cloud Services (24%)

The computing power platform for the orchard land- photovoltaic project.

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(III) Investment Plan

1. General Partner: Yichuan Investment has a registered capital of RMB 30 million and subscribes to 2% of the managed fund;

2. Dedicated Fund: Establish "Yichuan County Yunyan No. 1 Fund (Limited Partnership)", and the first phase is expected to raise CNY 1.5 billion, accounting for 20% of the total investment of CNY 7.5 billion in the first phase of the Yuanguang project. The sources of funds include Yichuan Asset Management's own capital (2%), China's state-owned enterprise mixed ownership reform fund, central state owned enterprise rural revitalization special fund, and Yan'an Revitalization and Development Industry Investment Fund, etc.;

3. Project loan: The remaining 80% of the funds are provided by policy banks.

China Commerce Puhui Holding Group Co., Ltd Beijing Jinlongli Investment Management Centre (Limited Partnership)

Jinlongli Enterprise Center (60%) Yichuan State-owned Assets Supervision and Administration Commission (40%) Jinlongli Enterprise Center (60%) Kai Gao (9%)

Yichuan County Urban Construction Investment Co., Ltd.

China Development Bank (56.5%) Yichuan State-owned Assets Supervision and Administration Commission (43.5%)

Yichuan Investment Management Co., Ltd.

China Commerce Puhui (40%) Jinlongli Investment (30%) Yichuan Investment (30%)

Chengtong Mixed-Ownership Reform Fund Central SOE Rural Revitalization Fund Yan'an Financial Development Fund

Yichuan County Yunyan No. 1 Fund (Limited Partnership)

Yichuan Investment (GP) Chengtong Mixed-Ownership Reform Fund (LP) Central SOE Rural Revitalization Fund (LP) China Development Bank Capital (LP)

Policy banks

Equity capital 20% Project financing 80%

Phase I of The computing power platform for the orchard land- photovoltaic project.

(CNY 7.5 billion)

(IV) Policy list

1. National policies and special support

• "National Agricultural Science and Technology Park" of the Ministry of Science and Technology;

• "Rural Comprehensive Reform Pilot Experiment" of the Ministry of Finance;

• "National Modern Agricultural Industrial Park" of the Ministry of Agriculture and Rural Affairs;

• Shaanxi node of the "Eastern Data and Western Computing Project" of the National Development and Reform Commission



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# 中商普惠控股集团投资参与设立陕西省延安市宜川县园光互补算力平台的方案

## 一、项目背景

### （一）政协立案

2023 年，煤矿文工团团长靳东委员提交《关于推进果树+光伏模式，助力乡村振兴》的提案，明确提出以矮砧密植苹果为主体，推动碲化镉等薄膜发电技术在林果产业中的融合应用；

### （二）延安示范

同年，延安市宜川县在 300 亩园光互补示范项目实现亩均产值 4.8 万元、年回报超 18%的基础上，宜川县与延安市两级政府高度重视，规划再推广 3000 亩，装机 150MWp，采用“矮砧密植果园+发电玻璃+微电网+集体分红”的一体化模式，将其列为重点支持工程；

### （三）发改委立项

2024 年 9 月 21 日，陕西省发改委印发《关于印发陕西省可再生能源发展三年行动方案（2024—2026）的通知》，**建设渭北 10 吉瓦级农光互补可再生能源基地；**

### （四）科技部重点

同年，“宜川园光互补”作为中欧联合实施的旗舰科研合作项目，纳入科技部“政府间国际科技创新合作”重点专项，该专项明确提出以光谱农业、薄膜发电与集体收益机制为基础，推动农光互补从“项目示范”走向“政策体系”奠定基础。

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二、项目概况

联合多家边缘计算、数据服务与AI企业，系统推进“算力+农业+能源”一体化方案设计，规划建设面向农业生产、绿色能源调度与AI智能系统的边缘算力平台。项目坚持“农业主导、能源协同、数字驱动”原则，由农业企业牵头提出场景需求，能源方提供低碳供能保障，数字方构建智能运维与AI模型，实现电力系统调度与农业数字化能力的深度融合。项目将有力撬动“双碳战略”与乡村振兴政策协同落地，形成可持续、可推广的“技术-金融-政策”一体化解决方案。

该项目已纳入科技部“政府间国际科技创新合作”重点专项，计划申报财政部“农村综合性改革试点”，并作为农业农村部国家级农业产业园申报主体。同时，项目正积极争取纳入国家发改委牵头推进的“东数西算”陕西枢纽节点建设计划。

**三、项目推进情况**

**（一）园光互补试点实施**

本项目自2023年5月1日在宜川县堡定村选择了300亩的果园进行本提案实验。种植矮砧树苗60000株，实验结果为全县种植最晚长势最好；每亩铺设碲化镉玻璃50%的面积，加上玻璃透光率20%，亩均得光率69%，实验结果为矮砧密植最佳光照比例；亩均发电装机52KWp，实验结果等同于集中式光伏。初步测试整体成功。2024年2月中商普惠控股集团与延安市相关领导进行会谈，初步确定本项目10000亩规模化实施选址宜川县云岩镇。

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**（二）园光互补算力中心规模化实施**

在园光互补试点全面成功的基础上，拟在陕西省延安市宜川县投资开发园光互补算力项目。项目一期总投资额为 **75 亿元人民币**，计划占地 1 万亩；引进矮化苗木 200 万株；铺设碲化镉光伏组件 600MWp（**15 亿**）；建设 500MW 级超大尺寸(1215×2300)薄膜发电玻璃生产线（**10 亿**）；建设 150MW 级算力中心（**45 亿**），并配备 300MW 功率、3 小时容量 900MWh 的储能系统（**5 亿**），以推动当地经济高质量发展。

**（三）实施建议**

经过前期多轮调研与协调，目前初步拟定由延安市宜川县政府引导平台中商普惠控股集团旗下园光互补产业平台公司——北京景龙农光科技有限公司，联合朗新科技集团股份有限公司旗下的分布式光伏聚合平台——新耀能源科技有限公司，以及由软通动力信息技术（集团）股份有限公司与中联云港数据科技股份有限公司合资设立的“东数西算”数据中心服务平台——北京软通云服信息科技有限公司，共同设立项目平台公司“宜川县中瓦园光数字科技有限公司”。

项目资金安排方面，拟由中商普惠组建的专项基金平台——宜川园光投资管理有限公司作为主要资金出资方参与投资，未来将结合各方职能定位及项目推进进度，进一步明确合作架构与权益分配机制。

## 四、投资方案及政策清单

**（一）中商普惠、景龙农光、宜川投资介绍**

中商普惠控股集团原名中商丝路控股集团，于 2022 年 2 月经中商控股集团授权设立并更名。鉴于本项目涉及土地流转、光伏指标等

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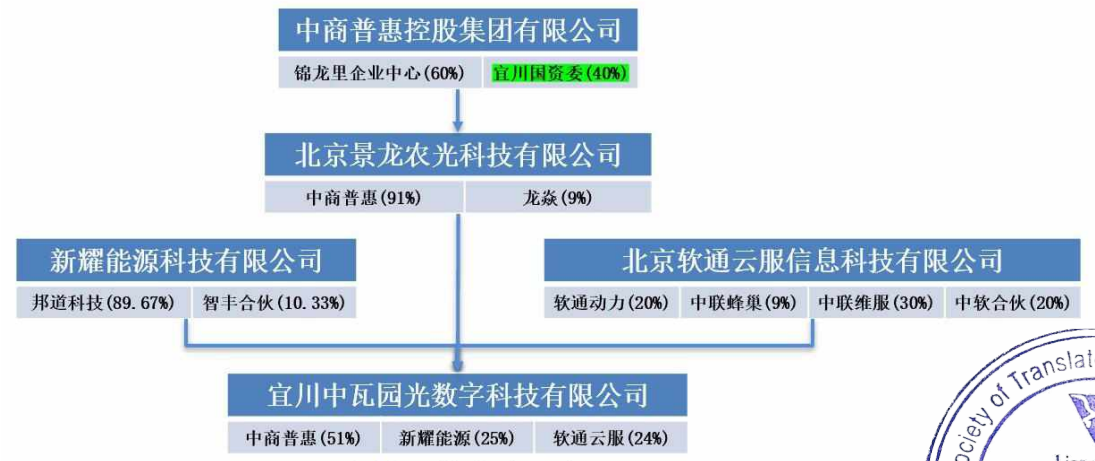
具体地方性事务，中商普惠已由项目所在地县政府牵头设立为园光互补产业的引导平台，具有明确的属地落地属性。该平台的设立，是延安市及宜川县政府共同推动的成果，也体现出地方政府对园光项目的高度重视和制度性支持。随着项目机制的成熟，未来将有更多具备条件的区县政府纳入该平台体系，共同推动园光互补模式的区域复制。

中商普惠设立产业平台北京景龙农光科技有限公司（简称景龙农光），股东单位龙焱能源科技（杭州）有限公司为碲化镉薄膜技术研发及生产世界头部企业。试点证实碲化镉光谱调控促进苹果生长。

中商普惠设立投资平台宜川园光投资管理公司（简称宜川投资），股东单位为国开发展基金有限公司与管理人团队北京锦龙里投资管理中心（有限合伙）。将募集管理园光互补算力产业相关基金。

**（二）开发主体**

拟由景龙农光、新耀能源科技有限公司及北京软通云服信息科技有限公司设立项目平台公司宜川县中瓦园光数字科技有限公司（简称中瓦数科，以工商注册登记核准为准），注册资本一亿元。其中景龙农光出资 5100 万占股 51%，新耀能源出资 2500 万占股 25%，软通云服出资 2400 万占股 24%，具体条件将在合资协议中约定。

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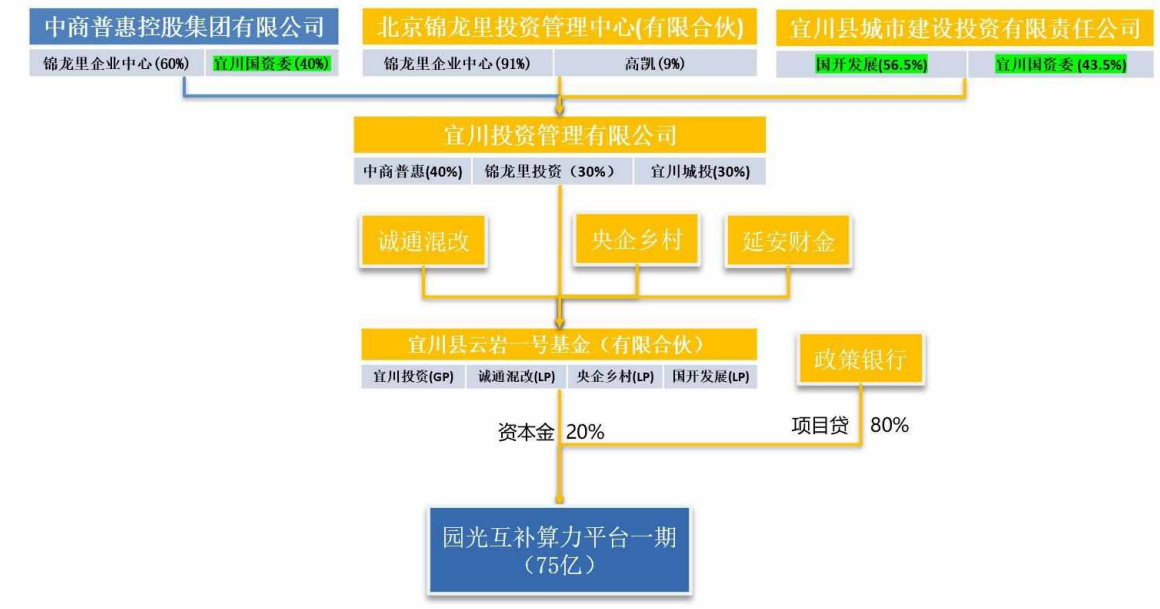
## （三）投资方案

1. 普通合伙人：宜川投资注册资本3000万元，认购所管基金2%；

2. 专项基金：设立“宜川县云岩一号基金（有限合伙）”，首期预计募集15亿元，对应一期园光项目总投资额75亿元的20%，资金来源包括宜川资管自有出资(2%)、中国国有企业混合所有制改革基金、央企乡村振兴专项基金以及延安振兴发展产业投资基金等；

3. 项目贷款：剩余80%资金由政策性银行提供项目贷款。




## （四）政策清单

1. 国家级政策与专项支持

- 科技部“国家级农业科技园”；
- 财政部“农村综合性改革试点试验”；
- 农业农村部“国家现代农业产业园”；
- 国家发展改革委“东数西算工程”陕西节点；

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